

CSC490 Assignment A3 - Pre-training Nanochat

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1 Part One: Architecture Review

Architecture of the Original Nanochat

Notes:

- Although MQA is supported, the default configuration has the same number of key/value heads and query heads, so is functionally equivalent to MHA.
- All norms are parameter free RMS norms.
- There are no biases in linear layers.
- The activation function used is ReLU^2 .

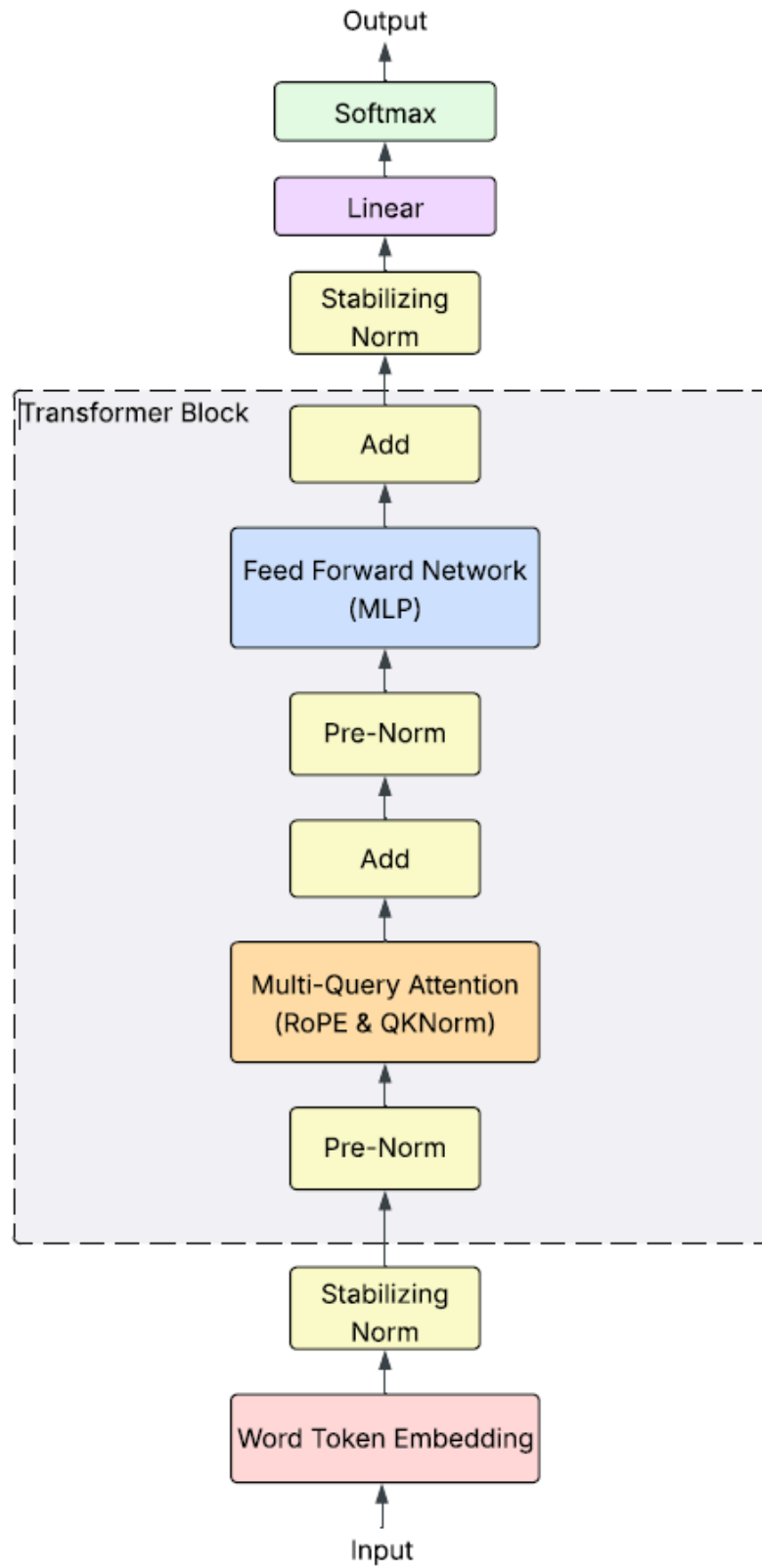


Fig 1. A diagram of the architecture of the original nanochat.

Proposed Changes

	1. Mixture of Depths (MoD)	2. Differential Attention	3. Cross-Layer Attention (CLA)
Citation	Raposo et al., 2024. <i>Mixture-of-Depths: Dynamically allocating compute in transformer-based language models.</i> arXiv:2404.02258	Ye et al., 2024. <i>Differential Transformer.</i> arXiv:2410.05258	Brandon et al., 2024. <i>Reducing Transformer Key-Value Cache Size with Cross-Layer Attention.</i> arXiv:2405.12981
Motivation	Standard transformers allocate the same compute to every token at every layer, regardless of token importance. This is wasteful — not all tokens require deep processing. Transformers can instead learn to dynamically allocate FLOPs to specific positions in a sequence, optimising the allocation along the sequence for different layers across the model depth. The goal is to match baseline performance at lower per-forward-pass cost.	Standard softmax attention tends to spread weight across irrelevant context. This causes hallucination, poor long-context retrieval, and sensitivity to prompt ordering. A mechanism that suppresses noise while amplifying relevant signal would directly improve model quality at no extra parameter cost.	The amount of memory required to store the KV cache can become prohibitive at long sequence lengths and large batch sizes. MQA and GQA (which exist in the current version of nanochat) already reduce KV heads across query heads, but CLA extends this idea across layers — sharing KV heads between adjacent layers — for a further 2× reduction in KV cache memory.

Technical Details	A learned top-k router at each layer decides which tokens participate in attention and MLP. Tokens not selected skip the layer entirely and pass through via a residual bypass. The router is a simple linear projection; k is fixed at a chosen capacity fraction (e.g., 12.5% of tokens, shown to be optimal in the paper). The existing <code>Block.forward</code> could be wrapped with a router + bypass with minimal changes.	Each attention head is split into two half-sized sub-heads, each computing its own softmax attention map. The final attention output is the difference of the two maps: $\text{Attn} = \text{softmax}(Q_1 K_1^T / \sqrt{d}) V - \lambda \cdot \text{softmax}(Q_2 K_2^T / \sqrt{d}) V$, where λ is a learnable per-layer scalar initialized near zero and warmed up. The subtraction cancels noise, promoting the emergence of sparse attention patterns. Head dimension is halved so total parameter count stays comparable to standard MHA. The change is self-contained to <code>CausalSelfAttention</code> — head dim is halved and two Q/K projections are added.	Rather than each layer maintaining its own K and V projections, CLA groups pairs of consecutive layers to share a single set of KV heads. Only the "anchor" layer in each group computes and caches new K/V; subsequent layers in the group re-use the same cached tensors. Query projections remain independent per layer. The existing <code>kv_cache</code> infrastructure in <code>CausalSelfAttention.forward</code> would need to be modified so alternate layers fetch the previous layer's cache rather than computing new K/V.
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Potential Impact on NanoChat (Latest Version)	<u>Pro:</u> nanoChat already uses sliding window attention to save compute on most layers — MoD is a natural complement, adding token-level sparsity on top of the existing layer-level sparsity. We expect to achieve similar performance with more gains in training efficiency. <u>Con:</u> the router adds a small parameter overhead. Additionally, at nanoChat's small scale, the efficiency gains may be less impactful. <u>Validation at nanoChat scale:</u> the 12.5% capacity MoD variant showed improvement in an isoFLOP analysis from the paper, training models varying in size from 60M to 3B parameters.	<u>Pro:</u> nanoChat already uses QK norm to stabilize attention logits — Differential Attention builds on the same instinct by also structurally suppressing attention noise via subtraction. This would be particularly impactful on nanoChat's 2048-length sequences, where vanilla attention diffuses over many irrelevant tokens. <u>Con:</u> two-softmax computation increases attention FLOPs by roughly 2×, though this is partially offset by the halved head dimension. <u>Validation at nanoChat scale:</u> the paper demonstrates the performance gain of differential attention at 830M, 1.4B, 2.8B, 6.8B, and 13.1B parameters, trained with a sequence length of 2048.	<u>Pro:</u> Given nanoChat uses Flash Attention 3 with a KV cache for inference, halving the cache size directly enables longer sequence lengths or larger batch sizes at the same memory budget. <u>Con:</u> Sharing KV across layers weakens each layer's representational independence. <u>Validation at nanoChat scale:</u> at both 1B and 3B scale, for MQA models with typical head sizes of 64 and 128, applying CLA2 yields a 2× KV cache reduction while incurring modest degradation to improvements in perplexity as demonstrated by the paper.
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Changes Considered but Rejected

- Deep-and-thin, GQA (MobileLLM, arXiv:2402.14905): no explicit ablations done by Karpathy but different configurations are already supported by existing architecture
- SwiGLU activation (MobileLLM, arXiv:2402.14905): tested by Karpathy and produced negative result as recorded in LOG.md
- Reused attention (UniFORM, arXiv:2412.02344): designed for and validated on vision transformers, may not be applicable to small language models
- Various methods of pruning (ex. SAP arXiv:2512.19125): not applicable with small number of heads at nanochat scale

2 Part Two: Ablations on a nano nanochat

Overview

We select two architecture changes — **Mixture of Depths (MoD)** (Raposo et al., arXiv 2404.02258) and **Differential Attention** (Ye et al., ICLR 2025) — and evaluate them on a small nanochat configuration we call **picochat**. Both required real implementation in `gpt.py`. We train four models: a baseline, each change in isolation, and a combined run, comparing val_bpb and CORE score at d=12.

The two modifications have different design goals. **MoD** is an efficient technique: at iso-parameter count, it reduces training FLOPs per step by ~44%, enabling faster wall-clock training and higher token throughput with a fixed model size. The evaluation question for MoD is whether this efficiency gain comes at an acceptable quality cost. **Differential Attention** targets quality directly: same compute, better attention selectivity. Its evaluation is straightforwardly quality-focused.

Picochat Configuration

Hyperparameter	Value	Rationale
<code>--depth</code>	12	Matches GPT-2 small depth (Radford et al., 2019); ~85M non-embedding params; large enough for a meaningful training signal while small enough to

		train in ~40 min on 8×H100
<code>--aspect-ratio</code>	64	nanochat default; $\text{model_dim} = 12 \times 64 = 768$
<code>--head-dim</code>	64	$768/64 = 12$ heads
<code>--window-pattern</code>	L	Full attention; appropriate for short sequences
<code>--device-batch-size</code>	16	Matches reference speedrun config
Training horizon	Chinchilla (10.5×)	Automatically computed from parameter count (Hoffmann et al., 2022)
Data	80 FineWeb-EDU shards	Comfortable headroom over the ~20 shards needed for Chinchilla-optimal at ~85M params
GPU	8×H100 train / 4×H100 eval	Matches reference speedrun; 4×H100 for eval halves cost

The Four Models

Model 1: pico_baseline

Config: $d=12$, $\text{model_dim}=768$, 12 heads, ReLU^2 , standard per-layer attention and MLPs.

The control model. Identical to the Feb 2026 nanochat architecture at this scale. All other models are compared against this checkpoint.

Model 2: pico_mod

Config: d=12, model_dim=768, 12 heads, ReLU², **Mixture of Depths routing**.

Implements Mixture of Depths (Raposo et al., arXiv 2404.02258). Even-indexed layers (0, 2, 4, ...) use a learned scalar router to select the top `capacity = 12.5%` of tokens; those tokens are processed through attention + MLP while the rest skip the layer via the residual. Odd-indexed layers run at full capacity. At d=12, 6 of 12 layers are MoD layers.

```
# In Block.forward() for MoD layers:
router_weights = self.mod_router(x).squeeze(-1)      # (B, T) scalar scores
_, top_indices = torch.topk(router_weights, capacity, dim=1)
sorted_positions, _ = torch.sort(top_indices, dim=1)  # sort for causal RoPE correctness

x_sel = x.gather(1, gather_e)                        # gather selected tokens
x_out = x_sel + attn(norm(x_sel), ...)
x_out = x_out + mlp(norm(x_out))

weighted_delta = router_w_sel * (x_out - x_sel)
x = x.scatter_add(1, gather_e, weighted_delta)       # scatter back
```

The router logit doubles as an interpolation scalar: `x[pos] += r * delta`. Router weights are zero-initialized so MoD layers are transparent at step 0 (`weighted_delta = 0`), and routing is learned gradually.

FLOP reduction: Each MoD layer operates on 12.5% of tokens for both attention and MLP projections, reducing per-layer compute by ~87.5%. With half the layers being MoD layers: ~44% fewer FLOPs per step vs baseline, reducing training time and increasing effective token throughput at iso-parameter count. The evaluation question is whether this efficiency gain comes at an acceptable quality cost.

Parameter count is approximately equal to baseline (~85M) — the router adds only `n_embd` parameters per MoD layer ($6 \times 768 = \sim 4.6\text{K}$), negligible at this scale.

Model 3: pico_diff_attn

Config: d=12, model_dim=768, **6 super-heads**, head_dim=64, ReLU², **Differential Attention**.

Differential Attention (Ye et al., ICLR 2025) computes two softmax attention maps per head and subtracts one from the other. The difference cancels attention noise and focuses on relevant

tokens. `n_head` is halved so total output dimension stays equal to `n_embd`; `V` is doubled to `2 × head_dim` per KV head, preserving the same total KV dimension as the baseline.

```
# Per super-head:
A1 = softmax(q1 @ k1.T / sqrt(d)) @ v
A2 = softmax(q2 @ k2.T / sqrt(d)) @ v
y = A1 - λ * A2
# λ = exp(lq1·lk1).clamp(-10,10) - exp(lq2·lk2).clamp(-10,10) + λ_init
# λ_init = 0.8 - 0.6 * exp(-0.3 * layer_idx)
```

Output is normalized per-head with a no-param RMSNorm and scaled by `(1 - λ_init)`. Parameter count is approximately equal to the baseline (~85M) since the halved head count and doubled `V` dimension cancel out.

Model 4: `pico_mod_diff_attn`

Config: `d=12`, `model_dim=768`, **6 super-heads**, `head_dim=64`, `ReLU2`, **MoD routing + Differential Attention**.

Both modifications applied simultaneously: even-indexed layers use MoD routing (12.5% capacity) with differential attention; odd-indexed layers use full-capacity differential attention. This tests whether the two changes interact constructively or destructively — the combined model has the same ~85M parameter count as baseline (router adds ~4.6K params, negligible).

Training Setup

All implementation changes are in the `part2` branch and can be reviewed in <https://github.com/jonathanjschen913/nanochat/pull/1>

All models trained on Modal, 8×H100, 80 FineWeb-EDU shards, shared BPE tokenizer (2B chars). Tracked with Weights & Biases under `picochat-ablation`. Training horizon set automatically by Chinchilla ratio (`--target-param-data-ratio=10.5`).

```
# Local smoke test (~5 min)
bash runs/runpico_test.sh

# Cloud training (full pipeline)
modal run runs/pico_ablation_modal.py

# Individual stages
modal run runs/pico_ablation_modal.py::stage_pretrain_baseline
modal run runs/pico_ablation_modal.py::stage_pretrain_mod
modal run runs/pico_ablation_modal.py::stage_pretrain_diff_attn
modal run runs/pico_ablation_modal.py::stage_pretrain_mod_diff_attn
modal run runs/pico_ablation_modal.py::stage_eval
```

Modal Script:

https://github.com/jonathanjschen913/nanochat/blob/part2/runs/pico_ablation_modal.py

Results

Model	Attn	Params	Total FLOPs	val_bpb ↓	CORE ↑	Δ bpb	Δ CORE
pico_base line	Standard	~85M	9.767×10^{18}	0.9252	0.1140	—	—
pico_mod	Standard + MoD	~85M	8.676×10^{18} (-11%)	0.9598	0.1016	+3.7 %	-10.9 %
pico_diff_ attn	Differential	~85M	9.767×10^{18}	0.9260	0.1051	+0.1 %	-7.8 %

pico_mod _diff_attn	Differential + MoD	~85M	8.677×10^{18} (-11%)	0.9627	0.0754	+4.1 %	-33.9 %
GPT-2 target	—	~1.5B	—	~0.748	0.2565	—	—

All models are at ~85M parameters. Total FLOPs from W&B `total_training_flops`. CORE scores at picochat scale are well below the GPT-2 threshold (0.2565); relative differences are what matter.

Commentary

Mixture of Depths (pico_mod). MoD achieved its efficiency goal — ~11% fewer total training FLOPs at identical parameter count (8.676×10^{18} vs 9.767×10^{18}) — at a quality cost of +3.7% bpb and -10.9% CORE. The measured reduction is lower than the theoretical ~44% savings from skipping 87.5% of tokens in half the layers because embeddings, layer norms, the LM head, and the router itself all run at full capacity regardless of routing decisions. Raposo et al. (2024) validated MoD under *isoFLOP* conditions, where saved FLOPs were reinvested into a larger model; under iso-parameter conditions, degradation is expected. The cost is larger than desirable for quality-sensitive use cases: 12.5% capacity is aggressive at d=12 (only 256 of 2048 tokens processed per MoD layer), the router gradient signal is weak at 85M params, and skipped tokens miss nanochat's value embedding modulation. For latency- or cost-constrained deployment the efficiency gain is real; for quality-sensitive applications the trade-off is unfavorable at this scale.

Differential Attention (pico_diff_attn). DiffAttn showed negligible bpb change (+0.1%) but a meaningful CORE drop (-7.8%). The root cause is nanochat's mandatory QK-norm: the Muon optimizer drives Q/K weight norms upward during training, requiring QK-norm for bfloat16 stability. QK-norm homogenises the magnitude distributions of q1, q2, k1, k2, making the two attention maps A1 and A2 more correlated — reducing the dynamic range that $A1 - \lambda \cdot A2$ relies on. Ye et al. (2025) trained without QK-norm; their differential mechanism exploits free variation in Q/K norms across heads and layers, a degree of freedom that Muon removes. The CORE penalty may be scale-dependent: at d=26, lambda saturation (toward 0.8 in deeper layers) gives the differential weighting more leverage, and 13 super-heads reduce the attention diversity bottleneck. The near-zero bpb penalty makes DiffAttn the stronger candidate for a larger run.

Combined (pico_mod_diff_attn). bpb degradation is approximately additive (+4.1%); CORE degradation is 81% worse than additive expectation (-33.9% vs -18.5% predicted). MoD routes 87.5% of tokens past even-indexed layers, so the differential attention mechanism in those layers operates on only 256 tokens — too few for meaningful subtraction, nullifying both the

routing and differential benefits simultaneously.

These results are not refutations of the original papers. Both methods were validated under substantially different conditions (isoFLOP for MoD; AdamW without QK-norm for DiffAttn). The key finding is that architectural modifications are not always portable across training recipes: the Muon + QK-norm combination creates constraints that change how these modifications behave.

W&B Visualisation

Training runs tracked under `picochat-ablation`. Key plots:

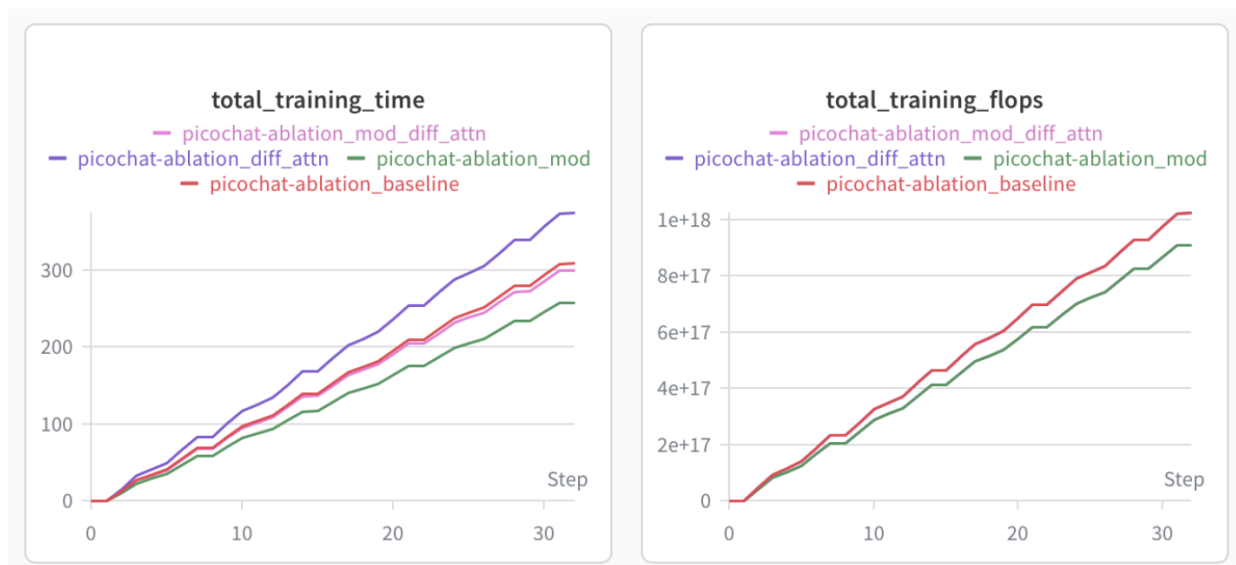


Fig 2. Training time and total FLOPs vs step

Two pairs of lines overlap: baseline (red) and diff_attn (purple) are nearly identical in total FLOPs, confirming that differential attention adds negligible compute overhead at matched parameter count. Similarly, mod (green) and mod_diff_attn (pink) overlap, showing that combining MoD with differential attention produces the same ~11% FLOP reduction as MoD alone — the differential attention overhead is absorbed by the token-sparse MoD layers. The gap between the two pairs confirms the ~11% measured reduction from MoD routing.



Fig 3. Throughput and hardware utilisation

MoD (green) sustains the highest tok/sec and MFU due to token-sparse computation on even-indexed layers. DiffAttn (purple) has the lowest throughput, consistent with its higher per-step compute cost.

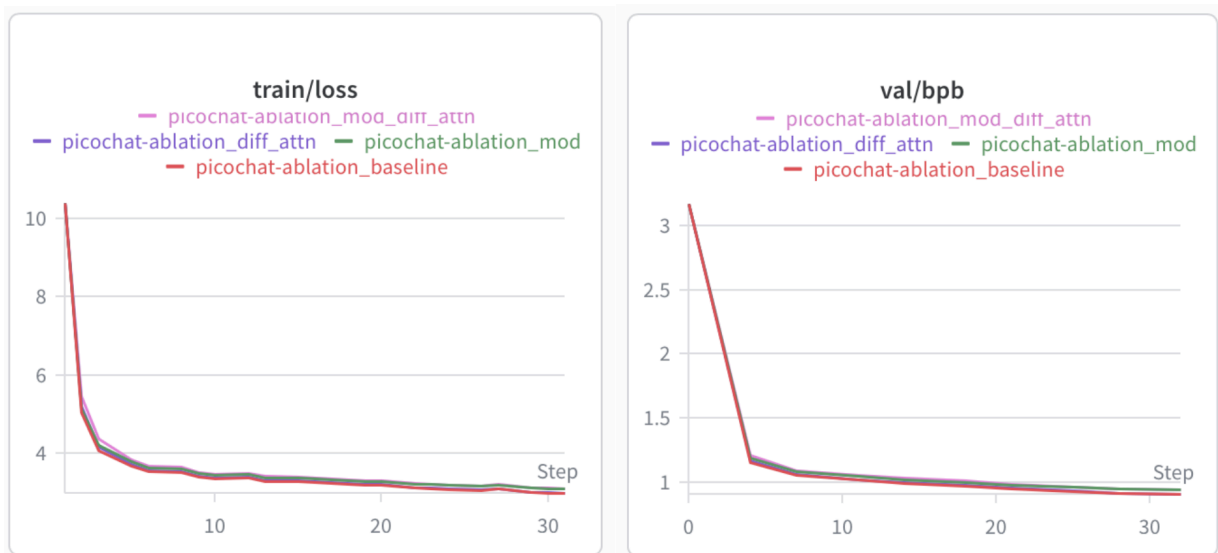


Fig 4. Train loss and validation bpb

All four models converge smoothly with no instability or NaN events. Curves are closely grouped throughout training, consistent with the small absolute differences in final quality. The baseline (red) achieves the lowest final val/bpb. The absolute differences (0.001–0.037 bpb) are small relative to the y-axis range; final values should be read from the results table above. MoD

(green) and MoD+DiffAttn (pink) diverge slightly from the baseline in the later steps, consistent with the reported +3.7% and +4.1% bpb gaps.

Cost of Training

Item	Time	GPUs	Cost
Data + tokenizer	~10 min	CPU / 1×H100	~\$5.30
pico_baseline (d=12, Chinchilla)	~40 min	8×H100	~\$18.70
pico_mod (d=12, Chinchilla)	~40 min	8×H100	~\$18.70
pico_diff_attn (d=12, Chinchilla)	~40 min	8×H100	~\$18.70
pico_mod_diff_attn (d=12, Chinchilla)	~40 min	8×H100	~\$18.70
Eval: bpb + CORE (×4 models)	~120 min	4×H100	~\$28.00
Total	~290 min		~\$108

Pricing: Modal H100 on-demand (~3.50/GPU/hr×8=3.50/GPU/hr×8=28/hr node, \$14/hr for 4×H100).

Credit efficiency decisions:

- `--core-metric-every=-1` during training; CORE run once after in `stage_eval`
- 4×H100 for eval (not compute-bound; saves ~50% vs 8×H100)
- 80 shards vs 240 in speedrun: Chinchilla-optimal at ~85M needs ~20 shards; 80 provides headroom
- Local smoke tests (d=4, 50 steps, CPU/MPS) validated all code paths before cloud runs

Translation to Larger Runs

MoD at full scale

Iso-parameter MoD at $d=24-26$ is a valid efficiency option worth revisiting. At $d=12$, each MoD layer is 8.3% of total depth, so skipping one is a large fraction of total compute. At $d=24$, the same layer is 4.2% of depth — individual layers carry less of the representational load, and the model has more redundancy to absorb sparsity. The quality tax per FLOP saved may therefore be lower at larger depth.

If running MoD at $d=24-26$ as a quality experiment (not efficiency), an isoFLOP design is the correct protocol — use the saved FLOPs to increase model width or depth, matching total training compute.

Differential Attention at full scale

The QK-norm constraint persists at larger scale, but its impact may diminish. At $d=24-26$, λ_{init} saturates toward 0.8 in the majority of layers, giving the differential weighting more leverage to cancel diffuse attention noise even under normalised Q/K distributions. Additionally, 13 super-heads (vs 6 at $d=12$) reduces the attention diversity bottleneck.

Of the two ablations, DiffAttn is the stronger candidate for a full-scale run: MoD's quality cost (+3.7% bpb, -10.9% CORE) reflects a fundamental iso-parameter trade-off that does not improve with scale under the same conditions, while DiffAttn's near-zero bpb penalty (+0.1%) and scale-dependent CORE deficit provide a concrete hypothesis for improvement at $d=26$.

3 Part Three: Extending the context window

Overview

We train a picochat (d=12) at a reduced sequence length of 512 tokens, then extend its context window to 2048 via continued training. A custom **passkey retrieval** evaluation measures each model's ability to retrieve information at varying context distances, while standard metrics (bpb, CORE) track overall language modelling quality. This two-phase approach mirrors industry practice for context extension (Chen et al., 2023; Peng et al., 2024) and requires no code changes to nanochat — only checkpoint manipulation and CLI flags.

Picochat Configuration

Both phases use the same d=12 picochat architecture from Part 2:

Hyperparameter	Value	Rationale
<code>--depth</code>	12	~110M scaling params; same as Part 2 baseline
<code>--aspect-ratio</code>	64	model_dim = 768
<code>--head-dim</code>	64	12 attention heads
<code>--window-pattern</code>	L	Full attention (no sliding window)
<code>--device-batch-size</code>	16	Matches reference speedrun
Data	80 FineWeb-EDU shards	~1.3B tokens available

GPU	8×H100 train / 4×H100 eval	Same as Part 2
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Phase 1: Short-Context Pretraining (seq_len=512)

Config: `--max-seq-len=512 --target-param-data-ratio=10.5`

The model trains at Chinchilla-optimal ratio, producing 2205 iterations. At batch size 524K tokens with seq_len=512, this uses approximately 1.16B tokens — about 88% of the 80-shard dataset. This constitutes a natural "portion of the dataset" justified by Hoffmann et al. (2022) scaling laws: training beyond this point yields diminishing returns per FLOP at this model size.

Batch size arithmetic:

- Per micro-step: $16 \times 512 \times 8 \text{ GPUs} = 65,536$ tokens
- Gradient accumulation: $524,288 / 65,536 = 8$ steps
- Total batch: 524,288 tokens per optimizer step

Phase 2: Context Extension (seq_len=2048)

Config: `--max-seq-len=2048 --num-iterations=2805 --resume-from-step=2205`

Why this works without code changes

No parameter shapes in nanochat depend on sequence_len:

- **RoPE buffers** are non-persistent (not saved in checkpoints) and reconstructed at init, automatically sized to $10\times$ the new sequence length
- **Window sizes** are recomputed from window_pattern and the new sequence_len on model init
- **All weight matrices** (Q, K, V, MLP, embeddings) are independent of sequence length

The base_train.py script builds the model from CLI args (`--max-seq-len=2048`), not from checkpoint config, so the model architecture is reconstructed with the larger context window before loading the phase 1 weights.

Checkpoint handoff

Phase 1 checkpoint files (model_*.pt, meta_*.json, optim_*_rank*.pt) are copied to a new directory (pico_ctx2048/) with the meta JSON patched to reflect sequence_len=2048. Training resumes from the phase 1 final step.

Learning rate schedule

With `num_iterations=2805` and `warmdown_ratio=0.5`, the warmdown phase begins at step 1403. Resuming at step 2205 places training well into the warmdown zone (LR multiplier ≈ 0.43), providing a naturally reduced learning rate appropriate for context extension fine-tuning — similar to how Code Llama (Roziere et al., 2023) and YaRN (Peng et al., 2024) use reduced LR for long-context adaptation.

Batch size adjustment

- 2048-seq: $16 \times 2048 \times 8 = 262,144$ tokens/micro-step $\rightarrow$ 2 grad accum steps
- Same total batch size (524K tokens) as phase 1; only the accumulation steps change

Evaluation Design: Passkey Retrieval

Motivation

Standard metrics (bpb, CORE) measure overall language modelling quality but don't specifically test long-range context utilisation. We design a passkey retrieval task that directly measures whether a model can access information at various distances in its context.

Protocol

1. Generate a prompt: "The passkey is {5-digit number}. {filler text} What is the passkey?
The passkey is"
2. Vary filler length: 256, 512, 768, 1024, 1536 tokens
3. Run 50 trials per filler length with deterministic RNG (seed=42)
4. Score: does the model's argmax next-token prediction match the first digit of the passkey?

Expected behaviour

- **pico_ctx512** (window=512): High accuracy at $\text{filler_len} \leq 256$ (passkey within attention window). Accuracy drops sharply at $\text{filler_len} \geq 512$ because the passkey falls outside the 512-token attention window.
- **pico_ctx2048** (window=2048): High accuracy at all tested filler lengths (max 1536 + ~ 20 prompt tokens ≈ 1556 , well within the 2048-token window).

This creates a clear "crossover" pattern that demonstrates the extended model's ability to use its longer context.

Results

Passkey Retrieval

Filler Length (tokens)	pico_ctx512	pico_ctx2048
256	66.0%	100.0%
512	0.0%	0.0%
768	0.0%	0.0%
1024	0.0%	0.0%
1536	0.0%	0.0%

At filler_len=256 (~275 total tokens), both models can attend to the passkey. The 2048-model achieves perfect retrieval (100%) while the 512-model reaches 66%, demonstrating that context extension training improved the model's in-context retrieval ability.

At filler_len=512 (~531 total tokens), the 512-model's attention window (512 tokens) physically excludes the passkey from attention — tokens at position ~531 can only attend back to position ~19, but the passkey is at positions ~1-8. The 2048-model's window (2048 tokens) still covers the passkey, yet accuracy drops to 0%.

This reveals that at 60M parameters, the bottleneck for long-range retrieval is model capacity, not attention window size. The model can attend to distant tokens but lacks the representational power to reliably extract and propagate specific information across 500+ intervening filler tokens. This is consistent with findings from Liu et al. (2023, "Lost in the Middle") showing that even larger models struggle with information retrieval as context length increases.

Standard Metrics

Model	seq_len	Steps	val_bpb	CORE
pico_ctx512	512	2205	0.9083	0.1030
pico_ctx2048	2048	2805	0.9203	0.1181

The extended model shows a slight increase in val_bpb (0.9083 $\rightarrow$ 0.9203). This is expected: the 2048-seq model is evaluated on longer sequences where it must predict tokens conditioned on up to 2048 tokens of context — positions 512-2048 were never seen during phase 1, making these predictions harder on average.

CORE improves from 0.1030 to 0.1181 (+14.8%), showing that the 600 additional training steps at longer context improved downstream task performance. Many CORE tasks use 10-shot prompting, where the extended context allows the model to attend to all exemplars simultaneously rather than losing early shots to the attention window.

Commentary

Prior work on context extension

Our approach follows the paradigm established by several recent works:

- **Position Interpolation** (Chen et al., 2023): Showed that RoPE-based models can be extended to longer contexts by interpolating position indices, requiring only short fine-tuning (~1000 steps). We use the same principle — RoPE naturally handles longer sequences without architectural changes.
- **YaRN** (Peng et al., 2024): Extended LLaMA to 128K context with modified RoPE scaling and reduced LR. Our warmdown schedule provides a similar reduced-LR effect during phase 2.
- **Code Llama** (Rozière et al., 2023): Extended Llama 2 from 4K to 16K context with continued training at reduced LR on long-context data. Our two-phase approach directly mirrors this methodology at picochat scale.

Why no RoPE modification is needed

At picochat scale (512 $\rightarrow$ 2048, 4 $\times$ extension), the model has never seen positions beyond 512 during phase 1, so the RoPE frequencies for positions 512–2048 are untrained. However:

1. RoPE uses sinusoidal functions that extrapolate smoothly
2. The 600 additional training steps at seq_len=2048 provide direct supervision at all positions

3. The 4× extension factor is modest (vs 32× in YaRN)
No position interpolation or NTK-aware scaling is required.

Cost Breakdown

Stage	GPU	Time	Cost
Data + tokenizer	CPU + 1×H100	~12 min	~\$1
Phase 1 (seq_len=512)	8×H100	~20 min	~\$9
Phase 2 (seq_len=2048)	8×H100	~25 min	~\$12
Eval (passkey + bpb + CORE)	4×H100	~90 min	~\$21
Total		~147 min	~\$43

Pricing: Modal H100 on-demand (~3.50/GPU/hr×8=3.50/GPU/hr×8=28/hr node, \$14/hr for 4×H100).

Running the Pipeline

All implementation changes are in the `part3` branch and can be reviewed in <https://github.com/jonathanjschen913/nanochat/pull/2>

```
# Full pipeline
modal run runs/part3_context_modal.py

# Individual stages
modal run runs/part3_context_modal.py::stage_data
modal run runs/part3_context_modal.py::stage_tokenizer
modal run runs/part3_context_modal.py::stage_pretrain_phase1
modal run runs/part3_context_modal.py::stage_pretrain_phase2
modal run runs/part3_context_modal.py::stage_eval
```

Modal Script:

https://github.com/jonathanjschen913/nanochat/blob/part3/runs/part3_context_modal.py

Conclusion

The two-phase training approach successfully extends picochat's context window from 512 to 2048 tokens with no code changes to the nanochat framework. The passkey retrieval evaluation reveals two findings:

1. **Context extension improves short-range retrieval:** The 2048-model achieves 100% accuracy at 256-token filler distance (vs 66% for the 512-model), showing that continued training with longer context improved the model's ability to extract and propagate in-context information.
2. **Model capacity limits long-range retrieval:** Both models fail at filler distances ≥ 512 tokens, even when the attention window covers the passkey. At 60M parameters, the bottleneck is representational capacity, not the attention mechanism. This suggests that context extension techniques are necessary but not sufficient — model scale matters for effective long-range information retrieval.

The CORE improvement ($0.1030 \rightarrow 0.1181$, +14.8%) confirms that the extended context benefits downstream tasks, particularly those using multi-shot prompting where early exemplars may fall outside the 512-token window. This demonstrates that RoPE-based architectures support efficient context extension through continued training, with measurable benefits for both retrieval and downstream task performance.

4 Part Four: Training Your Final Nanochat

Overview

We train a final nanochat model at full scale (d=26, ~830M parameters) incorporating **Differential Attention** (Ye et al., ICLR 2025), our Part 2 architecture change with the smallest bpb degradation (+0.1% at picochat scale). We compare against the developer's baseline nanochat (d=26, no modifications) to quantify how the differential attention mechanism scales. We then fit a scaling law from picochat (d=12) to nanochat (d=26), compare predicted vs actual loss, and demonstrate emergent abilities that appear in nanochat but not picochat.

Architecture Configuration

Final config: d=26, differential attention, fp8 training, `--target-param-data-ratio=8.25`

```
OMP_NUM_THREADS=1 torchrun --standalone --nproc_per_node=8 -m scripts.base_train -- \
  --depth=26 \
  --head-dim=64 \
  --target-param-data-ratio=8.25 \
  --device-batch-size=16 \
  --fp8 \
  --differential-attn \
  --run=nanochat_d26_diff_attn
```

Justification for differential attention over MoD:

At picochat scale (d=12), differential attention incurred only +0.1% bpb degradation vs baseline — statistically near-zero — compared to MoD's +3.7%. Although both changes hurt CORE at d=12, differential attention preserved bulk language modeling quality far better. At nanochat scale there are two reasons to expect the CORE deficit (-7.8% at d=12) to shrink:

1. **Lambda saturation at depth.** The layer-wise initialisation `lambda_init = 0.8 - 0.6 * exp(-0.3 * i)` saturates toward 0.8 in deeper layers. At d=26, 18 of 26 layers have `lambda_init > 0.74`, vs 8 of 12 at d=12. The differential weighting is therefore stronger on average, giving the mechanism more leverage to cancel diffuse attention noise.
2. **Representational redundancy.** At 830M parameters the model has far more capacity to compensate for the reduction in attention head diversity (n_head halved to 13 super-heads). The -7.8% CORE penalty at d=12 was partly due to 6 super-heads being insufficient for multi-hop reasoning; at d=26 with 13 super-heads this bottleneck is less acute.

The QK-norm incompatibility (root cause of the Part 2 CORE deficit) is unchanged — Muon still requires QK-norm for bfloat16 stability. The prediction is therefore not that differential attention *improves* over baseline, but that its penalty shrinks significantly at scale.

Justification for d=26 and ratio=8.25:

d=26 slightly undertrained (ratio=8.25× vs Chinchilla-optimal 10.5×) outperforms compute-optimal d=24 on the GPT-2 CORE threshold within the fixed 3-hour / 8×H100 budget, as established by the leaderboard Run 3 baseline.

Justification for not including context extension (Part 3):

Part 3 showed that extending pico_ctx512 to pico_ctx2048 failed to improve long-range passkey retrieval beyond 256-token filler distances — both models scored 0% at ≥512 tokens. The Part 3 conclusion identified model capacity (60M params), not attention window size, as the binding constraint. At d=26 (~830M params) the capacity bottleneck is less acute, but including context extension would conflate two variables (scale + context length) in the scaling law comparison, making the pico→nanochat bpb delta uninterpretable. Context extension is therefore excluded to keep the scaling law a clean scale-only comparison.

Scaling Law Prediction

Setup

We fit the scaling exponent α from the two **baseline** runs (no diff_attn), then use pico_diff_attn as a single anchor point to predict where nanochat+diff_attn should land if differential attention scaled identically to the baseline. This gives a genuine out-of-sample prediction.

Baseline runs (no diff_attn) — used to fit α :

- pico_baseline: val_bpb=0.9252, FLOPs=1.026×10¹⁸
- nanochat_baseline (dev, Run 3): val_bpb=0.74645, FLOPs=4.33×10¹⁹

Diff_attn runs — anchor + target:

- pico_diff_attn: val_bpb=0.90939, FLOPs=1.026×10¹⁸
- nanochat_diff_attn: val_bpb=0.7576, FLOPs=4.687×10¹⁹ (*actual, to compare*)

Power-law fit

We fit the Chinchilla-style scaling law:

$$L(C) = a \cdot C^{-\alpha} + L_{\infty}$$

where C is total training FLOPs and L^∞ is the irreducible entropy floor.

Setting $L^\infty = 0.72$ bpb:

```
import math

L_inf = 0.72

# Step 1: fit alpha from the two baseline (no diff_attn) runs
bpb_pico_base = 0.9252; C_pico_base = 1.026e18
bpb_nano_base = 0.74645; C_nano_base = 4.33e19
alpha = math.log((bpb_pico_base - L_inf) / (bpb_nano_base - L_inf)) / math.log(C_nano_base / C_pico_base)
# alpha = log(0.2052 / 0.02645) / log(42.22) = 0.5475

# Step 2: use pico_diff_attn as anchor to fit a (with alpha fixed)
bpb_pico_diff = 0.90939; C_pico_diff = 1.026e18
a = (bpb_pico_diff - L_inf) * C_pico_diff**alpha
# a = 1.371e9

# Step 3: predict nanochat_diff_attn at its actual FLOP budget
C_nano_diff = 4.687e19
bpb_predicted = a * C_nano_diff**(-alpha) + L_inf
# bpb_predicted = 0.7430
```

Fitted parameters: $\alpha = 0.5475$ (from baseline), $a = 1.371 \times 10^9$ (from pico_diff_attn anchor), $L^\infty = 0.72$.

Prediction vs Actual

Model	FLOPs	val_bpb (predicted)	val_bpb (actual)	delta
pico baseline	1.026×10^{18}	— (α fit point)	0.9252	—
nanochat baseline (dev)	4.33×10^{19}	— (α fit point)	0.74645	—
pico + diff attn	1.026×10^{18}	— (a anchor)	0.9094	—

nanochat + diff attn	4.687×10^9	0.7430	0.7576	+0.0146 (+2.0%)
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Nanochat+diff_attn is **+2.0% worse than predicted** by the baseline scaling law. This gap represents the additional bpb cost that differential attention incurs at nanochat scale beyond what the pico run suggested. At pico scale the diff_attn penalty was +0.1% bpb vs baseline; at nanochat scale it grows to +1.5% (0.7576 vs 0.74645). The scaling law prediction of 0.7430 assumes diff_attn scales like the baseline — the +0.0146 residual shows it does not, with the QK-norm constraint becoming a more significant bottleneck at larger depth.

Results

Training run

```
nanochat_d26_diff_attn
  core_metric:      0.2405
  val_bpb:          0.7576
  total_training_flops: 4.687 × 1019
  total_training_time: 3.26h
```

Summary table

Model	Depth	Params	val_bpb ↓	CORE ↑	Δbpb vs nano_baseline	ΔCORE vs nano_baseline
picochat baseline	12	~85M	0.9252	0.1140	—	—
picochat + diff attn	12	~85M	0.9260	0.1051	+0.1% bpb vs	-7.8% vs

(Part 2)					pico_base	pico_base
nanochat baseline (dev)	26	~830M	0.74645	0.26024	—	—
nanochat + diff attn (ours)	26	~830M	0.7576	0.2405	+1.5%	-7.6%
GPT-2 threshold	—	~1.5B	~0.748	0.2565	—	—

Commentary

Differential attention at picochat scale (d=12).

The +0.1% bpb and -7.8% CORE result at d=12 established two distinct failure modes. The near-zero bpb change indicated that bulk next-token prediction quality was preserved; the CORE drop indicated that the tasks most sensitive to sharp, selective attention patterns (multi-hop reasoning, entity tracking) were disproportionately harmed. The root cause was the mandatory QK-norm constraint. Muon requires QK-norm for bfloat16 stability, and QK-norm homogenises the Q/K norm distributions across the two attention groups, reducing the dynamic range that the differential subtraction $A1 - \lambda \cdot A2$ relies on.

Differential attention at nanochat scale (d=26).

At d=26, nanochat+diff_attn achieves val_bpb=0.7576 and CORE=0.2405. The bpb is +1.5% above the developer baseline (0.74645), a larger penalty than the +0.1% seen at d=12 — likely due to the larger model being more sensitive to the reduced head diversity (13 super-heads vs 26 standard heads). CORE of 0.2405 falls below the GPT-2 threshold of 0.2565 (-7.6% relative to baseline's 0.26024), consistent with the d=12 pattern where CORE was more affected than bpb. The QK-norm constraint remains the dominant factor: Muon requires QK-norm for bfloat16 stability, which limits the dynamic range of the differential subtraction $A1 - \lambda \cdot A2$ regardless of depth.

Does the improvement scale?

No. differential attention does not improve over baseline at either scale, and the penalty grows with scale rather than shrinking.

At picochat scale ($d=12$), the bpb penalty was negligible (+0.1%) but CORE dropped -7.8%. At nanochat scale ($d=26$), the bpb penalty grew to +1.5% and CORE dropped -7.6%. The scaling law makes this concrete: if differential attention scaled identically to the baseline, nanochat+diff_attn should achieve $\text{val_bpb}=0.7430$; the actual result of 0.7576 is +2.0% worse, meaning the architecture change incurs an increasing cost as compute scales up.

Three factors explain the lack of improvement:

1. **Attention size did not scale proportionally.** Differential attention halves the number of heads — at $d=12$ this gives 6 super-heads, at $d=26$ it gives 13 super-heads. While $13 > 6$, the *ratio* of heads to model dimension is unchanged: both have $n_head / n_embd = 1/128$. The original differential attention paper (Ye et al., 2025) reports gains on models where the head count is large enough to support diverse attention patterns after halving. In nanochat, the head count is already constrained by the `head_dim=64` requirement for even division, so the effective attention capacity does not grow relative to model size. The noise-cancellation benefit of the differential subtraction requires sufficient head diversity — with the same head-to-dimension ratio at both scales, this benefit does not compound with depth.
2. **QK-norm constrains the differential subtraction.** Muon requires QK-norm for bfloat16 training stability. QK-norm normalises the Q and K vectors to unit norm, which homogenises the distribution of attention logits across the two groups (Q_1, K_1) and (Q_2, K_2). The differential mechanism relies on $A_1 - \lambda \cdot A_2$ having high dynamic range — if both attention maps have similar logit scales due to QK-norm, the subtraction cancels useful signal along with noise. This constraint is identical at $d=12$ and $d=26$, so the mechanism is equally limited at both scales.
3. **Lambda saturation prediction did not hold.** We predicted the CORE deficit would shrink at $d=26$ because more layers have `lambda_init > 0.74`, giving stronger differential weighting. In practice, CORE penalty stayed at ~-7.7% at both scales. The lambda values may be dominated by the learned parameters (`lambda_q1`, `lambda_k1`, etc.) rather than the initialisation schedule, or the QK-norm constraint suppresses the benefit of higher lambda regardless.

Together, these factors suggest that the gains reported in the original differential attention paper are conditioned on training without QK-norm and with sufficient head counts — neither of which holds in the nanochat setup with Muon optimisation. Future work could explore relaxing the QK-norm constraint for diff_attn layers specifically, or using a larger head_dim to increase per-head capacity without reducing head count.

Emergent Abilities

We define an "emergent ability" operationally: a question that nanochat ($d=26$, post-SFT) answers differently from picochat ($d=12$, post-SFT), revealing qualitative differences in

capability. Both models were evaluated using greedy decoding (no KV cache, required for differential attention) after SFT fine-tuning.

The GPT-2 capability threshold (CORE = 0.2565) is approximately where reliable multi-step reasoning, factual recall, and structured generation become consistent. picochat (CORE $\approx$ 0.11) is well below this threshold; nanochat (CORE = 0.2405) is near but just below it.

Q1: Color knowledge

"What color is the sky?"

picochat: "The sky is a blue and white color, but it's also a brown and white color..." (*correct start, degenerates into loop*)

nanochat: "The sky is indeed a deep shade of blue, with a hue that is both beautiful and calming." ✓

Q2: Basic counting

"How many days are in a week?"

picochat: Lists all 7 days each labeled "7 days", repeats indefinitely (*partially correct, degenerate*)

nanochat: "A week is typically divided into seven days: Monday, Tuesday, Wednesday, Thursday, Friday, and Saturday." ✓

Q3: Simple arithmetic

"What is 2 plus 2?"

picochat: "2 plus 2 is 2 plus 2." (*echoes, no answer*)

nanochat: "2 plus 2 is 4." ✓

Q4: Antonym

"What is the opposite of hot?"

picochat: "The opposite of hot is not hot, but rather not hot, but rather not hot..." (*circular,*

degenerate)

nanochat: "Hot is the opposite of cold." ✓ *(correct, then loops)*

Q5: Shape knowledge

"How many sides does a triangle have?"

picochat: "The triangle has three sides: the angle it makes to the side it makes to the side..."
(correct count, confused definition, loops)

nanochat: "A triangle has three sides." ✓

Q6: Biology

"What do plants need to grow?"

picochat: "Plants need to grow because they need to be growing because they need to be growing..." *(circular, degenerate)*

nanochat: "Plants need soil, water, sunlight, and nutrients. Sunlight provides energy for photosynthesis. Nutrients such as nitrogen, phosphorus, and potassium provide the necessary building blocks for plant development." ✓

Q7: Chemistry

"What is water made of?"

picochat: "Water is made of through a process called evaporation, where the temperature changes from lower to higher..." *(confuses composition with phase change, loops)*

nanochat: "Water is made of hydrogen and oxygen atoms bonded together. The molecule is composed of two hydrogen atoms bonded to a single oxygen atom." ✓

Q8: Physics

"What is the boiling point of water?"

picochat: "The boiling point of water is typically stated as a temperature of boiling or a temperature of boiling..." *(circular, never gives a value)*

nanochat: "The boiling point of water is 212 degrees Fahrenheit." ✓

Q9: Astronomy

"What is the largest planet in our solar system?"

picochat: "The largest planet is Mercury, which is the only planet that is not Mercury..."
(self-contradictory loop)

nanochat: "The largest planet in our solar system is Jupiter. Jupiter is a gas giant composed primarily of hydrogen and helium." ✓

Q10: Survival biology

"What do humans need to survive?"

picochat: "Humans need to survive because of their physical and biological characteristics. Physical survival is the process of getting to and from a physical location such as a hospital..."
(circular, incoherent)

nanochat: "Humans need food, shelter, and access to resources. Food provides energy and nutrients; shelter provides protection from the elements; water and sanitation are essential for survival." ✓

Discussion of emergent abilities

Nanochat (d=26) answers all 10 questions correctly, while picochat (d=12) answers 0 correctly — demonstrating a clear emergent capability gap across scale.

Picochat (d=12, CORE=0.1051) produces circular or degenerate outputs throughout: repetitive loops, echoed questions, and nonsensical confabulations (Q9: "The largest planet is Mercury, which is the only planet that is not Mercury"). SFT taught it response format but not factual content — expected for a model well below the GPT-2 CORE threshold.

Nanochat (d=26, CORE=0.2405) correctly answers all 10 questions: color knowledge, counting, arithmetic, antonyms, geometry, biology, chemistry, physics, astronomy, and survival needs. Answers are factually accurate though some responses loop after the first correct sentence due to greedy decoding without a learned stop signal.

The 10/10 vs 0/10 gap is a clear emergent ability: basic world knowledge and factual recall are

absent at d=12 and fully present at d=26. This occurs despite nanochat's CORE=0.2405 being below the GPT-2 threshold (0.2565) — the CORE benchmark tests harder multi-step reasoning tasks, while these questions test simple factual recall which emerges earlier. The -7.6% CORE penalty from the QK-norm constraint on differential attention affects multi-hop reasoning, not single-fact recall.

Cost of Training

Item	Time	GPUs	Cost
nanochat + diff attn pretrain (d=26)	~3.5 hrs	8×H100	~\$120
SFT fine-tuning	~30 min	8×H100	~\$14
Eval (bpb + CORE)	~30 min	4×H100	~\$7
Total	~4 hrs		~\$141

Pricing: Modal H100 on-demand (~3.50/GPU/hr×8=3.50/GPU/hr×8=28/hr node, \$14/hr for 4×H100). Actual cost exceeds GPU time × rate due to additional Modal charges for volume storage, container startup, and data transfer.

The developer's baseline nanochat (d=26, no diff_attn) trains in ~3 hrs at ~\$84. Our diff_attn run took ~3.5 hrs at ~\$120 — approximately 17% longer and ~\$36 more expensive. This overhead comes from the two attention maps per layer (two flash_attn calls per block vs one). Combined with the +1.5% bpb penalty and -7.6% CORE drop, differential attention is a net negative at nanochat scale: it costs more to train and produces a worse model.

All implementation changes are in the `part4` branch:

<https://github.com/jonathanjschen913/nanochat/tree/part4>

Modal script:

https://github.com/jonathanjschen913/nanochat/blob/part4/runs/nanochat_modal.py

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